

Exercise 1: Per-unit values

A system has $S_{base} = 100 \text{ MVA}$

- Calculate the base current for $V_{base} = 525 \text{ kV}$
- Using this value, calculate the actual line current and phase voltage where $I = 4.95 \text{ p.u.}$, and $U = 0.5 \text{ p.u.}$

$$I_{base} = \frac{1000 \times 100}{\sqrt{3} \times 525} \text{ amperes} = 110.0 \text{ A}$$

$$I_{actual} = (4.95) \cdot (110.0 \text{ A}) = 544 \text{ A}$$

$$V_{actual} = (0.5) \cdot (525 \text{ kV}) = 263 \text{ kV}$$

(Marx 2012)

Exercise 2: Per-unit of transformer impedance

A 900 MVA, 525/241.5 autotransformer has a nameplate impedance of 10.14%

- a) Determine the impedance in ohms, referenced to the 525 kV side
- b) Determine the impedance in ohms, referenced to the 241.5 kV side

First convert from % to *pu*.

$$Z_{pu} = \frac{Z\%}{100} = 0.1014$$

$$Z(\Omega) = Z_{pu} \frac{kV_{base}^2}{MVA_{base}}; \text{ therefore}$$

$$\begin{aligned} \text{a) } Z_{525kV} &= 0.1014 \times \frac{525^2}{900} \\ &= 31.05\Omega \end{aligned}$$

$$\begin{aligned} \text{b) } Z_{241.5kV} &= 0.1014 \times \frac{241.5^2}{900} \\ &= 6.57\Omega \end{aligned}$$

(Marx 2012)

Exercise 3: Faults and symmetrical components

Given $Z_0 = 0.199\angle 90^\circ$ p.u., $Z_1 = 0.175\angle 90^\circ$ p.u., $Z_2 = 0.175\angle 90^\circ$ p.u.,

- compute the fault current and voltages for a single line-to-ground fault
- compute the fault current and voltages for a line-to-line fault
- compute the fault current and voltages for a double line-to-ground fault

For the solution, refer to examples 8.2 through 8.4 (Marx 2012):

Example 8.2

Given $Z_0 = 0.199\angle 90^\circ$ pu, $Z_1 = 0.175\angle 90^\circ$ pu,

$Z_2 = 0.175\angle 90^\circ$ pu, compute the fault current and voltages for a Single line-to-ground fault. Note that the sequence impedances are in *per-unit*. This means that the results for current and voltage will be in *per-unit*.

Solution

The sequence networks are interconnected in series, as shown in Fig. 8.5

Because the sequence currents are in series, and using ohms law,

$$I_0 = I_1 = I_2$$

$$I_0 = \frac{V_1}{(Z_0 + Z_1 + Z_2)}$$

$$\begin{aligned} I_0 &= \frac{1\angle 0^\circ}{(j0.199 + j0.175 + j0.175)} \\ &= -j1.82 \text{ pu} \end{aligned}$$

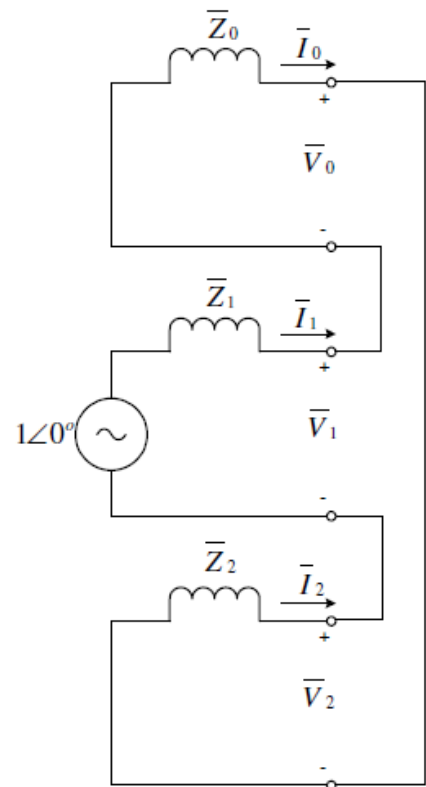


Fig 8.5

The phase currents are converted from the sequence value using Eq. (7.8). Substituting $I_0 = I_1 = I_2$ into Eq. (7.8) gives

$$I_a = I_0 + I_0 + I_0 = 3I_0$$

$$I_b = I_0 + a^2 I_0 + a I_0 = 0$$

$$I_c = I_0 + a I_0 + a^2 I_0 = 0$$

Refer to Table 3.2: $(1 + a + a^2 = 0)$

Note that $I_a = 3I_0$. This is the quantity that the relay “see’s” for a Single Line-to-Ground fault.

Substituting $I_0 = -j1.82 pu$

$$\begin{aligned} I_a &= 3I_0 = 3(-j1.82) \\ &= -j5.46 pu \end{aligned}$$

Calculating the voltage drop, and referring to Fig. 8.5, the sequence voltages are

$$V_0 = -Z_0 I_0$$

$$V_1 = V - Z_1 I_1$$

$$V_2 = -Z_2 I_2$$

Substituting in the impedance and current from above

$$V_0 = -j0.199(-j1.82) = -0.362$$

$$V_1 = 1 - j0.175(-j1.82) = 0.681$$

$$V_2 = -j0.175(-j1.82) = -0.319$$

The phase voltages are converted from the sequence value using Eq. (7.6).

$$V_a = -0.362 + 0.681 - 0.319 = 0$$

$$V_b = -0.362 + a^2(0.681) + a(-0.319) = 1.022 \angle 238^\circ pu$$

$$V_c = -0.362 + a(0.681) + a^2(-0.319) = 1.022 \angle 122^\circ pu$$



Fig 8.6

The *per-unit* value for the current and voltage would now be converted to actual values using Eq. (5.9) and Eq. (5.10) and knowing the base power and voltage for the given system. See example 5.1 for a reference.

The currents and voltages can be shown in phasor form in Fig. 8.6

Example 8.3

Given $Z_0 = 0.199\angle 90^\circ pu$, $Z_1 = 0.175\angle 90^\circ pu$, $Z_2 = 0.175\angle 90^\circ pu$, compute the fault current and voltages for a Line-to-Line fault. Note that the sequence impedances are in *per-unit*. This means that the solution for current and voltage will be in *per-unit*.

Solution

The sequence networks are interconnected, as shown in Fig. 8.7

Because the sequence currents sum to one node, it follows that

$$I_1 = -I_2$$

The current I_1 is the voltage drop across Z_1 in series with Z_2

$$\begin{aligned} I_1 &= \frac{V_1}{Z_1 + Z_2} \\ I_1 &= \frac{1\angle 0^\circ}{j0.175 + j0.175} \\ &= -j2.86 pu \end{aligned}$$

$$I_2 = +j2.86 pu$$

$$I_0 = 0$$

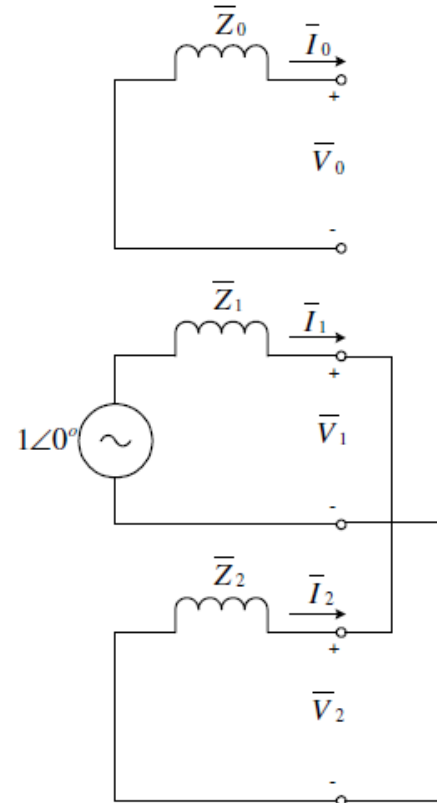


Fig 8.7

The phase current is converted from the sequence value using Eq. (7.8).

$$I_a = 0 - j2.86 + j2.86 = 0 pu$$

$$I_b = 0 + a^2(-j2.86) + a(j2.86) = -4.95 pu$$

$$I_c = 0 + a(-j2.86) + a^2(j2.86) = 4.95 pu$$

Calculating the voltage drop, and referring to Fig. 8.7, the sequence voltages are

$$V_1 = V_2$$

$$\begin{aligned} V_2 &= -Z_2 I_2 \\ &= -(j1.75)(j2.86) \\ &= 0.5 pu \end{aligned}$$

$$V_0 = 0$$

The phase voltages are converted from the sequence value using Eq. (7.6).

$$V_a = 0.0 + 0.5 + 0.5 = 1.0 \text{ pu}$$

$$V_b = 0.0 + a^2(0.5) + a(0.5) = -0.5 \text{ pu}$$

$$V_c = 0.0 + a(0.5) + a^2(0.5) = -0.5 \text{ pu}$$

The *per-unit* value for the current and voltage would now be converted to actual values using Eq. (5.9) and Eq. (5.10) and knowing the base power and voltage for the given system. See example 5.1 for a reference.

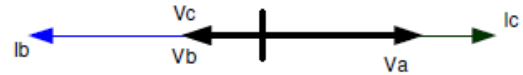


Fig (8.8)

The currents and voltages can be shown in phasor form in Fig. 8.8

Example 8.4

Given $Z_0 = 0.199\angle 90^\circ pu$, $Z_1 = 0.175\angle 90^\circ pu$, $Z_2 = 0.175\angle 90^\circ pu$, compute the fault current and voltages for a Double Line-to-Ground fault. Note that the sequence impedances are in *per-unit*. This means that the solution for current and voltage will be in *per-unit*.

Solution

The sequence networks are interconnected, as shown in Fig. 8.9

Because the sequence currents sum to one node, it follows that

$$I_1 = -(I_0 + I_2)$$

The current I_1 is the voltage drop across Z_1 in series with the parallel combination of Z_0 and Z_2

$$I_1 = \frac{V_1}{Z_1 + \left(\frac{Z_0 Z_2}{Z_0 + Z_2} \right)}$$

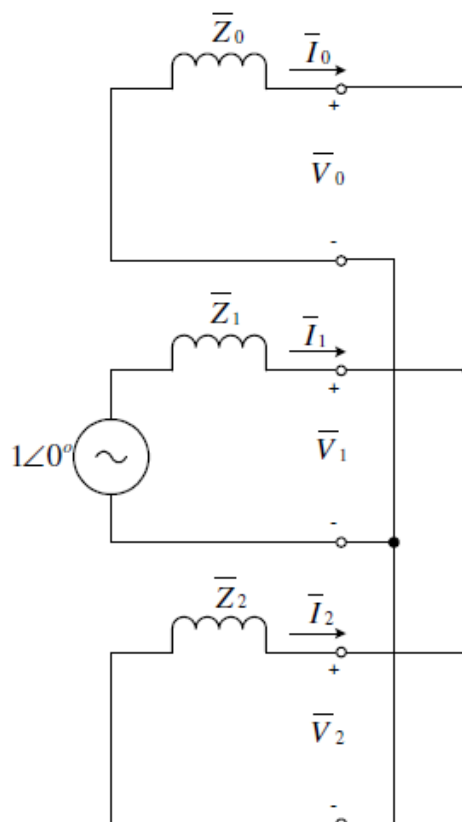


Fig 8.9

Substituting in $V_1 = 1\angle 0^\circ$, and Z_0 , Z_1 , and Z_2 , then solving for I_1

$$I_1 = -j3.73 pu$$

$$I_0 = \frac{Z_2}{(Z_0 + Z_2)} I_1$$

$$= +j1.75$$

$$I_2 = \frac{Z_0}{(Z_0 + Z_2)} I_1$$

$$= +j1.99$$

The phase current is converted from the sequence value using Eq. (7.8).

$$I_a = j1.75 - j3.73 + j1.99 = 0 pu$$

$$I_b = j1.75 + a^2(-j3.73) + a(j1.99) = 5.60\angle 152.1^\circ pu$$

$$I_c = j1.75 + a(-j3.73) + a^2(j1.99) = 5.60\angle 27.9^\circ pu$$

Calculating the voltage drop, and referring to Fig. 8.9, the sequence voltages are

$$V_0 = V_1 = V_2$$

$$V_0 = -Z_0 I_0$$

$$= -(j1.75)(j0.199)$$

$$= 0.348 pu$$

The phase voltages are converted from the sequence value using Eq. (7.6).

$$V_a = 0.348 + 0.348 + 0.348 = 1.044 pu$$

$$V_b = 0.348 + a^2(0.348) + a(0.348) = 0 pu$$

$$V_c = 0.348 + a(0.348) + a^2(0.348) = 0 pu$$

$$\text{Refer to Table 3.2: } (1 + a + a^2 = 0)$$

The *per-unit* value for the current and voltage would now be converted to actual values using Eq. (5.9) and Eq. (5.10) and knowing the base power and voltage for the given system. See example 5.1 for a reference.

The currents and voltages can be shown in phasor form in Fig. 8.10

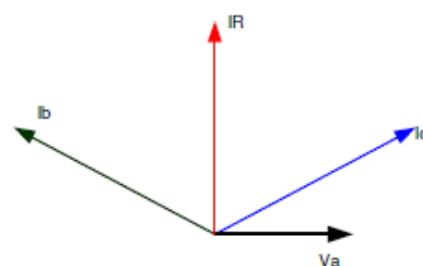


Fig 8.10

Exercise 4: Faults and symmetrical components

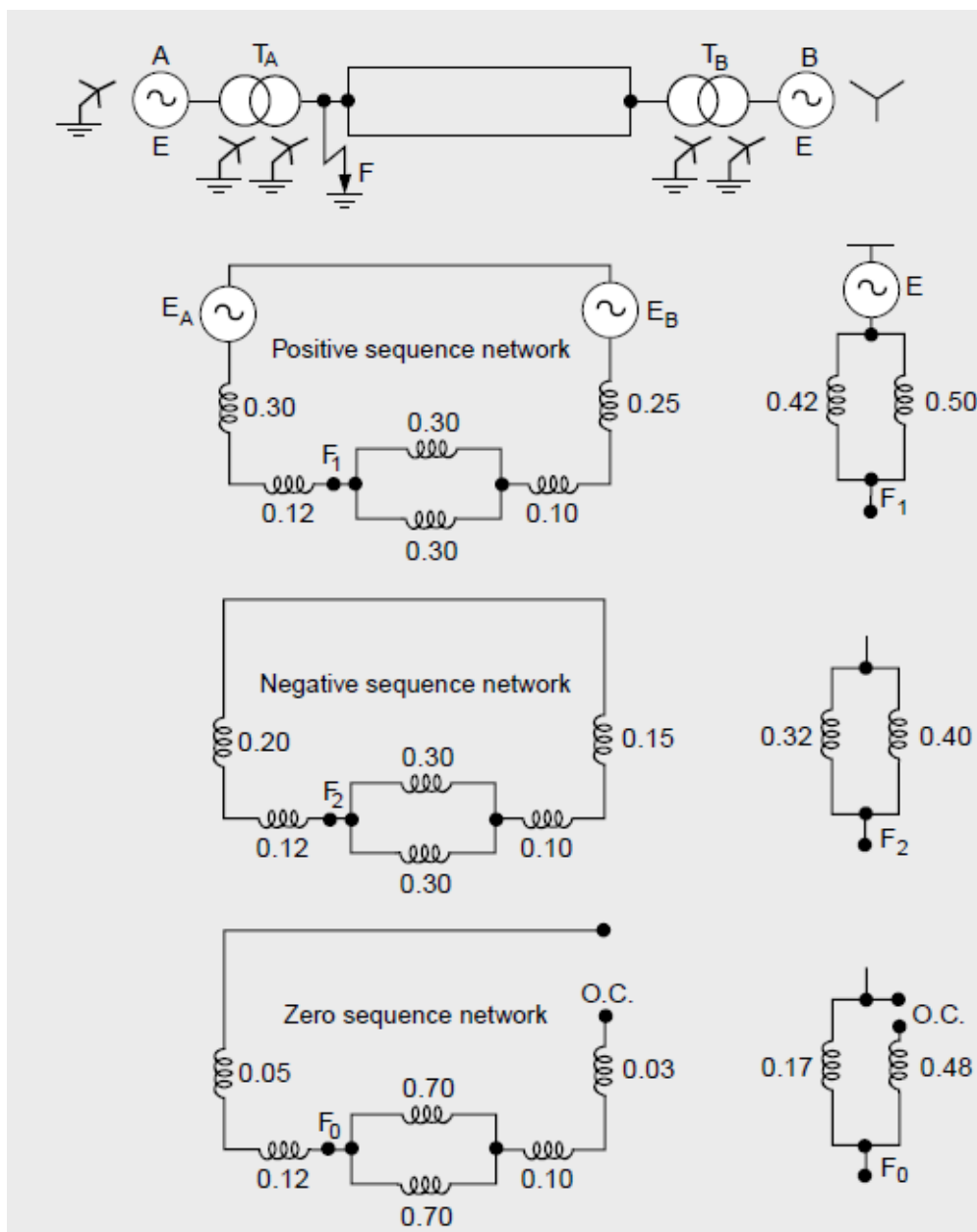
A synchronous machine A generating 1 p.u. voltage is connected through a star-star transformer, reactance 0.12 p.u., to two lines in parallel. The other ends of the lines are connected through a star-star transformer of reactance 0.1 p.u. to a second machine B, also generating 1 p.u. voltage. For both transformers, $X_1 = X_2 = X_0$.

The relevant per unit reactances of the plant, all referred to the same base, are as follows:

- For each line: $X_1 = X_2 = 0.3$, $X_0 = 0.7$
- For machine A: $X_1 = 0.3$, $X_2 = 0.2$, $X_0 = 0.05$
- For machine B: $X_1 = 0.25$, $X_2 = 0.15$, $X_0 = 0.03$.

The star points of machine A and of the two transformers are solidly earthed.

(Sequence networks are depicted in the image below.)



Calculate the current fed into a double line-to-earth fault on the line-side terminals of the transformer fed from A.

Solution

The positive-, negative-, and zero-sequence networks are shown in Figure 7.21. All per unit reactances are on the same base. From these diagrams the following equivalent reactances up to the point of the fault are obtained: $Z_1 = j0.23$ p.u. $Z_2 = j0.18$ and $Z_0 = j0.17$ p.u.

The red phase is taken as reference phasor and the blue and yellow phases are assumed to be shorted at the fault point. From the equivalent circuit for a line-to-line fault,

$$\begin{aligned} I_1 &= \frac{E}{Z_1 + [Z_2 Z_0 / (Z_2 + Z_0)]} \\ &= \frac{1}{j[0.23 + [0.17 \times 0.18 / (0.17 + 0.18)]]} = -j3.15 \text{ p.u.} \end{aligned}$$

$$\begin{aligned} I_2 &= -I_1 \frac{Z_0}{Z_2 + Z_0} \\ &= -j3.15 \times \frac{0.17}{0.17 + 0.18} = -j1.53 \text{ p.u.} \end{aligned}$$

$$\begin{aligned} I_0 &= -I_1 \frac{Z_2}{Z_2 + Z_0} \\ &= -j3.15 \times \frac{0.18}{0.17 + 0.18} = -j1.62 \text{ p.u.} \end{aligned}$$

$$\begin{aligned} I_Y &= I_0 + a^2 I_1 + a I_2 \\ &= j1.62 + (-0.5 - j0.866)(-j3.15) + (-0.5 + j0.866)(j1.53) \\ &= -4.05 + j2.43 \text{ p.u.} \end{aligned}$$

$$\begin{aligned} I_B &= I_0 + a I_1 + a^2 I_2 \\ &= j1.62 + (-0.5 + j0.866)(-j3.15) + (-0.5 - j0.866)(j1.53) \\ &= 4.05 + j2.43 \text{ p.u.} \end{aligned}$$

$$I_Y = I_B = 4.72 \text{ p.u.}$$

The correctness of the first part of the solution can be checked as

$$\begin{aligned} I_R &= I_0 + I_1 + I_2 \\ &= j1.62 - j3.15 + j1.53 = 0 \end{aligned}$$

(Weedy 2012)